

Global bank: project and delivery

01

Lead Data Engineer (contract) London, UK, November 2025 to present

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Global banking data platform
Trade-finance ELT for risk analytics and regulatory reporting.
Data: transactions, counterparty hierarchies, credit limits, collateral and exposure.
Consumers: Risk and Regulatory Reporting.

Team and responsibilities

8 people: 4 data engineers, 1 architect, 1 BA, 1 SM and 1 DM. Two data engineers report to me.

Lead design reviews on performance, reliability, security and cloud cost. Guide implementation and reusable patterns.

Publish reconciled data across ingestion, transformation and publication.

Agile delivery process

Suggested interview walkthrough

1. Backlog refinement and planning
Clarify requirements, acceptance criteria and dependencies.
 2. Implementation and review
Build pipelines, review code and validate rules.
 3. Validation and release
Reconcile data, demonstrate results and deploy through CI/CD.
 4. Retrospective and improvement
Review issues and agree actions for the next cycle.
- To confirm: sprint length, ceremonies, delivery tool and release cadence.

Scope: trade-finance operations, risk analytics and regulatory reporting. Agile cadence remains to confirm.

Batch ingestion: LC and parties

02

Proposed: F = full refresh; I = watermark + MERGE; C = scheduled CDC; A = immutable append. Initial snapshot for all tables.

PARTY [C]

PK party_id
name, country, customer_ref
party_type, risk_rating
Buyer / importer or seller / exporter

BANK [F]

PK bank_id
BIC, bank_name, country
branch_ref
Issuing / advising / confirming roles

ACCOUNT [C]

PK account_id
FK party_id, bank_id
account_ref, currency, type
status (customer / nostro)

TRADE_CONTRACT [I]

PK contract_id
FK buyer_id, seller_id (PARTY)
contract_ref, incoterm
amount, currency, signed_date

CONTRACT_ITEM [I]

PK item_id; FK contract_id
goods_description, HS_code
quantity, unit, unit_price
country_of_origin

LC_APPLICATION [C]

PK application_id
FK contract_id, applicant_id
requested_amount, currency
submitted_at, approval_status

LETTER_OF_CREDIT [C]

PK lc_id; FK application_id
lc_number, amount, currency
issue_date, expiry_date
payment_terms, status

LC_AMENDMENT [I]

PK amendment_id; FK lc_id
version_no, effective_at
changed_terms, approval_status
amount_delta, expiry_change

LC_PARTY_ROLE [C]

PK (lc_id, party_id, role)
FK lc_id, party_id
role: applicant / beneficiary
valid_from, valid_to

LC_BANK_ROLE [C]

PK (lc_id, bank_id, role)
FK lc_id, bank_id
role: issuing / advising / confirming
confirmation_status, effective_at

LC_DOC_REQUIREMENT [I]

PK requirement_id; FK lc_id
document_type, required_copies
required_terms, due_date
lc_version_no

SHIPMENT [C]

PK shipment_id
FK lc_id, contract_id
carrier, ports, shipped_at
shipment_ref, delivery_status

Source DB / files → ADF Copy → ADLS Gen2 Bronze → Databricks PySpark / Delta → Silver / Gold

Full refresh: BANK + country, currency, product, status and document-type reference feeds. FX history loads incrementally. Mutable PARTY / ACCOUNT use CDC.

All rows: source_system, source_record_id, source_updated_at, operation, ingested_at, batch_id. Partition Bronze by source / table / ingestion date. Keep raw history and reconcile source totals.

Batch ingestion: settlement and risk

03

Proposed: F = full refresh; I = watermark + MERGE; C = scheduled CDC; A = immutable append. Initial snapshot for all tables.

DOC_PRESENTATION [C]

PK presentation_id; FK lc_id
FK presenter_id (PARTY)
presented_at, presentation_status
claimed_amount, currency

DOCUMENT_METADATA [I]

PK document_id
FK presentation_id, shipment_id
FK requirement_id; type, doc_ref
storage_uri, hash, received_at

DOCUMENT_EXAM [I]

PK exam_id
FK presentation_id, bank_id
examined_at, result
examiner_ref, rule_version

DISCREPANCY [C]

PK discrepancy_id
FK exam_id, document_id
code, detail, decision
waived_by, resolved_at

SETTLEMENT [C]

PK settlement_id; FK lc_id
FK presentation_id,
debit_account_id
FK credit_account_id; amount, ccy
value_date, status, payment_ref

FEE_CHARGE [C]

PK fee_id; FK lc_id
FK settlement_id, account_id
fee_type, amount, currency
charged_at, charge_status

SWIFT_MESSAGE [C]

PK message_id; FK lc_id
FK sender_bank_id,
receiver_bank_id
message_type, direction, sent_at
message_ref, payload_uri, status

CREDIT_FACILITY [C]

PK facility_id; FK party_id
limit_amount, currency
valid_from, valid_to, status
LC links via FACILITY_USAGE

FACILITY_USAGE [A]

PK (facility_id, lc_id, as_of_ts)
FK facility_id, lc_id
utilised_amount, available_amount
currency, source_event_ref

COLLATERAL_POSITION [C]

PK collateral_id; FK party_id
asset_type, value, currency
valuation_at, haircut_pct
LC allocation needs bridge table

LC_COLLATERAL [C]

PK (lc_id, collateral_id)
FK lc_id, collateral_id
allocated_value, currency
allocation_at, release_at

LC_EXPOSURE_EVENT [A]

PK event_id; FK lc_id
exposure_amount, currency
exposure_type, lifecycle_status
occurred_at, source_event_ref

Source DB / files → ADF Copy → ADLS Gen2 Bronze → Databricks PySpark / Delta → Silver / Gold

CDC: read change-feed offsets and apply inserts / updates / deletes. Incremental: use reliable update timestamps, commit watermarks after success; reconcile hard deletes.

Document / SWIFT payloads: secure object storage; ingest metadata + URI. Platform controls: ingestion_config, watermark_control, batch_audit, schema_registry, reconciliation_control.

Source systems and related tables

Trade-finance application / outbox
LC issue / amendment, presentation, discrepancy and shipment events.

Tables: LETTER_OF_CREDIT, LC_AMENDMENT, DOC_PRESENTATION, DISCREPANCY, SHIPMENT.

Payment service / SWIFT gateway
Payment and message status events.

Tables: SETTLEMENT, FEE_CHARGE, SWIFT_MESSAGE. The bank gateway publishes events.

Risk / collateral services
Exposure, facility usage and valuation changes.
Tables: LC_EXPOSURE_EVENT, FACILITY_USAGE, COLLATERAL_POSITION, LC_COLLATERAL.

Database CDC connector
Publishes row changes when applications lack events. CDC capture can also feed scheduled batch.

Event contract: event_id, lc_id, event_type, event_time, schema_version, source_system, source_offset. Partition by lc_id where order matters. Immutable events append; mutable state uses MERGE.

Proposed mapping. CV: Kafka streaming. Architecture: Event Hubs. Confirm broker topology and actual event producers.

Ingestion service and destination

PROPOSED STREAMING ROUTE

Application outbox / CDC connector → Kafka topics OR Azure Event Hubs → Databricks Structured Streaming → ADLS Gen2 Bronze Delta → Silver / Gold Delta.

ADF orchestrates batch ingestion. It is not the primary streaming consumer in this design.

Kafka / Event Hubs: durable event transport. Event Hubs supports a Kafka endpoint; confirm the actual topology.

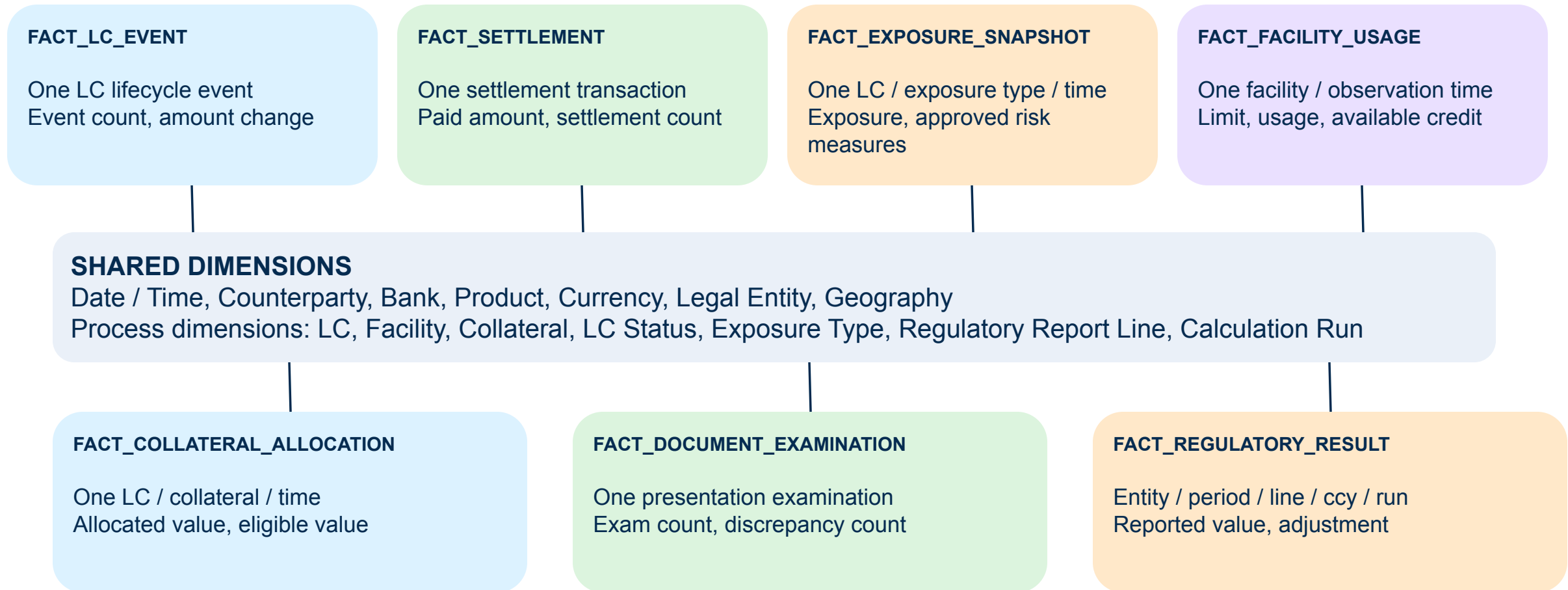
Databricks: schema validation, event-time handling, deduplication and checkpointing. Use replay-safe, idempotent writes and source offsets.

Batch reconciliation: compare source records / totals and repair gaps. Reconcile stream events with batch snapshots by source keys to avoid double counting.

Global bank: consumer galaxy

05

Seven fact tables share conformed dimensions. Each fact keeps its own grain and uses the applicable dimensions.



Proposed model, not verified Global bank DDL. Trade-finance scope; additional marts and source feeds are needed for other bank-wide risks.

Consumer facts: grains and measures

06

FACT_LC_EVENT

Grain: one immutable LC lifecycle event
PK lc_event_sk; source event_id
FK lc_sk, counterparty_sk, bank_sk
FK date/time, product, currency, entity
Measures: event_count = 1,
amount_delta
Attributes: event_type, lifecycle_status

FACT_SETTLEMENT

Grain: one settlement transaction
PK settlement_sk; source settlement_id
FK lc_sk, payer/payee counterparty_sk
FK debit/credit bank, date, ccy, entity
Measures: paid_amount,
transaction_count
Attributes: settlement_status,
payment_ref

FACT_EXPOSURE_SNAPSHOT

Grain: LC × exposure type × as_of_time
PK exposure_sk; unique business grain
FK lc, counterparty, product, ccy, entity
FK exposure_type, as_of_date/time
Measures: outstanding_exposure
PD/LGD/EAD need approved risk feeds

FACT_FACILITY_USAGE

Grain: facility × observation timestamp
PK usage_sk; aggregate LC-level
source
FK facility, counterparty, date/time, ccy
Measures: limit, utilised, available
Store facility total once per snapshot
Utilisation ratio = utilised / limit

FACT_COLLATERAL_ALLOCATION

Grain: LC × collateral × as_of_time
PK allocation_sk; version / source key
FK lc, collateral, counterparty, ccy, date
Measures: allocated_value,
eligible_value
Keep allocation history and haircuts
Avoid repeating collateral market value

FACT_DOCUMENT_EXAMINATION

Grain: one exam of a document
presentation
PK examination_sk; source exam_id
FK lc, bank, counterparty, exam_date
Measures: exam_count,
discrepancy_count
Attributes: result, rule_version
Bridge exam to documents as needed

Date/Time and Bank can play different roles. Currency conversions use the rate date and an approved FX feed. Exposure, limits and collateral are snapshot measures; do not sum them across time.

FACT_REGULATORY_RESULT

Grain: reporting entity × period × report line × currency × calculation run

PK regulatory_result_sk

FK legal_entity_sk, period_date_sk, report_line_sk, currency_sk, calculation_run_sk

Measures: reported_value, adjustment_value

Context: framework/version, rule version, source cutoff, approval status

Dimension and relationship rules

Dimension PK to fact FK uses 1:N relationships.

Use conformed identifiers and business definitions across facts.

Track relevant Counterparty / Facility history with SCD Type 2.

Facts reference the version effective at the event or snapshot time.

Use bridges for LC bank/party roles, exposure allocation and report contributions where relationships are many-to-many.

Reporting and aggregation controls

Aggregate each fact to the shared report grain before combining results. Direct fact-to-fact joins can multiply amounts.

Reconcile to source / ledger control totals. Retain calculation versions, lineage and approval evidence.

BRIDGE_REPORT_CONTRIBUTION maps a result to contributing source/fact records and allocation weights where needed.

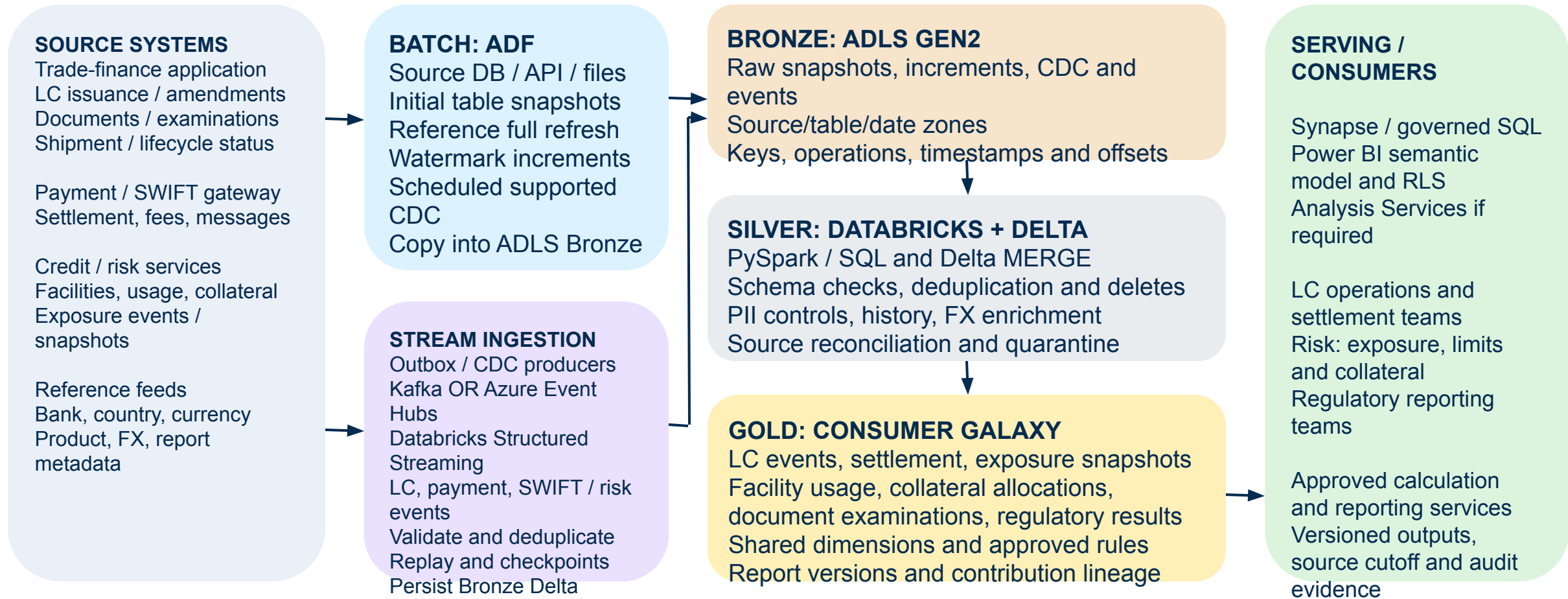
Coverage: LC operations, settlement, trade-finance exposure, facility usage, collateral and regulatory outputs.

Bank-wide market, liquidity, operational risk and capital reporting need additional facts, approved calculation engines and risk/reference feeds.

Global bank: Azure architecture

08

Proposed design: source systems → ingestion → ADLS / Databricks → consumer galaxy → reporting



Batch: Sources → ADF → Bronze → Databricks Silver / Gold → SQL / Power BI

Streaming: Producers / CDC → Kafka OR Event Hubs → Structured Streaming → Bronze → Silver / Gold

PLATFORM: Unity Catalog / Purview, Key Vault, scoped access, monitoring, Terraform and CI/CD. ADF / Airflow orchestrate batch. Reconcile batch and streams to prevent gaps and double counting.

1. Source contracts and batch

ADF loads the 24 proposed LC tables using initial snapshots, reference full refreshes, update-timestamp increments or scheduled supported CDC. Preserve source keys, operations and offsets. Commit watermarks after success and reconcile deletes.

3. Bronze and Silver processing

ADLS Bronze preserves raw history. Databricks Silver checks schemas, deduplicates, handles deletes and applies ordered Delta MERGE. Protect PII, quarantine failures, retain required history and enrich with approved reference / FX feeds.

5. Reporting and regulatory evidence

Synapse / governed SQL and Power BI serve operations and risk teams with consistent semantics and RLS. Regulatory results retain report-line grain, rule/run versions, source cutoff and approvals. Reconcile source/ledger totals and preserve contribution lineage.

2. Streaming producers and transport

Trade-finance, payment/SWIFT gateway and risk services publish through an outbox; CDC connectors can publish row changes. Kafka OR Event Hubs transports events. Databricks Structured Streaming consumes them. ADF is not the streaming consumer.

4. Gold galaxy and calculations

Seven facts retain distinct grains and share conformed dimensions. Snapshot measures do not sum across time. Aggregate facts to a shared report grain before combining. Approved risk feeds / engines supply missing PD, LGD and EAD measures.

6. Reliability and release controls

Checkpoint streams, deduplicate by event identity and use replay-safe writes. Reconcile streams with batch by source keys. Monitor freshness, failures and quality. Key Vault and catalog / identity controls protect data; Terraform and CI/CD manage releases.